

1)

Find the SQL query that corresponds to the following query execution plan:

Finalize Aggregate (cost=247198.03..247198.04 rows=1 width=8)

Output: count(*)

Gather (cost=247197.71..247198.02 rows=3 width=8)

Output: (PARTIAL count(*))

Workers Planned: 3

Partial Aggregate (cost=246198.03..246198.04 rows=1 width=8)

Output: PARTIAL count(*)

Parallel Hash Join (cost=201282.25..245999.89 rows=79129 width=0)

Hash Cond: (mk.movie_id = t.id)

-> Parallel Seq Scan on public.movie_keyword mk

(cost=0.00..39047.32 rows=1459332 width=4)

Output: mk.id, mk.movie_id, mk.keyword_id

-> Parallel Hash (cost=200853.84..200853.84 rows=34273 width=16)

Output: t.id, mi.movie_id, mi_idx.movie_id, mc.movie_id

-> Nested Loop (cost=17512.26..200853.84 rows=34273 width=16)

Output: t.id, mi.movie_id, mi_idx.movie_id, mc.movie_id

-> Nested Loop (cost=17511.83..179506.97 rows=33212 width=12)

Output: t.id, mi.movie_id, mi_idx.movie_id

-> Parallel Hash Join (cost=17511.39..62964.41 rows=62564 width=8)

Output: t.id, mi_idx.movie_id

Hash Cond: (t.id = mi_idx.movie_id)

-> Parallel Seq Scan on public.title t

(cost=0.00..43898.97 rows=344896 width=4)

Output: t.id, t.title, t.imdb_index, t.kind_id,
t.production_year, t.imdb_id, t.phonetic_code,
t.episode_of_id, t.season_nr, t.episode_nr,
t.series_years, t.md5sum

Filter: (t.production_year > 2000)

-> Parallel Hash (cost=15122.68..15122.68 rows=191097 width=4)

Output: mi_idx.movie_id

-> Parallel Seq Scan on public.movie_info_idx mi_idx

(cost=0.00..15122.68 rows=191097 width=4)

Output: mi_idx.movie_id

Filter: (mi_idx.info_type_id = 101)

-> Index Scan using movie_id_movie_info on public.movie_info mi

(cost=0.43..1.82 rows=4 width=4)

Output: mi.id, mi.movie_id, mi.info_type_id, mi.info, mi.note

Index Cond: (t.id = mi.movie_id)

Filter: (mi.info_type_id = 8)

-> Index Only Scan using movie_id_movie_companies on public.movie_companies mc

(cost=0.43..0.59 rows=5 width=4)

Output: mc.movie_id

Index Cond: (mc.movie_id = t.id)

Answer

```
SELECT COUNT(*)
FROM title AS t
JOIN movie_info_idx AS mi_idx
    ON t.id = mi_idx.movie_id
JOIN movie_info AS mi
    ON t.id = mi.movie_id
JOIN movie_companies AS mc
    ON t.id = mc.movie_id
JOIN movie_keyword AS mk
    ON t.id = mk.movie_id
WHERE t.production_year > 2000
    AND mi_idx.info_type_id = 101
    AND mi.info_type_id = 8;
```

The query returns:

26718423

This query matches the given execution plan. The plan performs COUNT(*) over the result of joining title, movie_info_idx, movie_info, movie_companies, and movie_keyword. The filters are t.production_year > 2000, mi_idx.info_type_id = 101, and mi.info_type_id = 8. The joins are performed through title.id = movie_id for each related table.

2)

Optimize the following query.

Try different join orderings or different operators. Explain your solution and provide the execution plans before and after the optimization in JSON format.

Before running the query, force PostgreSQL to respect the written join order:

Option 1 — using psql:

1. ALTER ROLE your_username SET join_collapse_limit = 1;

2. Quit psql and reconnect to the database:

```
\q
```

3. Check the setting:

```
SHOW join_collapse_limit;
```

The value must be 1.

Option 2 — using pgAdmin:

Add the following command as the first row before the query:

```
SET join_collapse_limit = 1;
```

Query:

```
SELECT COUNT(*)
```

```
FROM title t,
```

```
    movie_info mi,
```

```
    movie_companies mc,
```

```
    cast_info ci,
```

```
    movie_keyword mk
```

```
WHERE t.id = mi.movie_id
```

```
    AND t.id = mc.movie_id
```

```
    AND t.id = ci.movie_id
```

```
    AND t.id = mk.movie_id
```

```
    AND ci.role_id = 2
```

```
    AND mi.info_type_id = 16
```

```
    AND t.production_year > 2000
```

```
    AND t.production_year < 2010
```

AND mk.keyword_id = 7084;

Answer

I forced PostgreSQL to respect the written join order by setting:

SET join_collapse_limit = 1;

SHOW join_collapse_limit;

The value returned was 1.

Original query:

```
SELECT COUNT(*)
FROM title t,
      movie_info mi,
      movie_companies mc,
      cast_info ci,
      movie_keyword mk
WHERE t.id = mi.movie_id
      AND t.id = mc.movie_id
      AND t.id = ci.movie_id
      AND t.id = mk.movie_id
      AND ci.role_id = 2
      AND mi.info_type_id = 16
      AND t.production_year > 2000
      AND t.production_year < 2010
      AND mk.keyword_id = 7084;
```

Optimized query:

```
SELECT COUNT(*)
FROM movie_keyword mk
JOIN title t
    ON t.id = mk.movie_id
JOIN cast_info ci
    ON ci.movie_id = t.id
JOIN movie_info mi
    ON mi.movie_id = t.id
JOIN movie_companies mc
    ON mc.movie_id = t.id
WHERE mk.keyword_id = 7084
    AND t.production_year > 2000
    AND t.production_year < 2010
    AND ci.role_id = 2
    AND mi.info_type_id = 16;
```

Explanation:

The optimized query starts from movie_keyword using the selective predicate mk.keyword_id = 7084, which returns 414 rows. Then it joins title and applies the production_year filter, reducing the intermediate result to 199 rows. After that, it joins cast_info with ci.role_id = 2 and movie_info with mi.info_type_id = 16. The unfiltered movie_companies table is joined last, because it can multiply the number of rows and should not be joined early.

The final number of joined rows before aggregation is the same in both plans, 1067496, so the result is unchanged. The execution time decreased from 1682.153 ms to 232.171 ms. Shared hit blocks decreased from 13392145 to 166857. Therefore, the new join ordering is substantially more efficient.

Execution plan before optimization in JSON format:

```
[
{
  "Plan": {
    "Node Type": "Aggregate",
    "Strategy": "Plain",
    "Partial Mode": "Simple",
    "Parallel Aware": false,
    "Async Capable": false,
    "Startup Cost": 3280.24,
    "Total Cost": 3280.25,
    "Plan Rows": 1,
    "Plan Width": 8,
    "Actual Startup Time": 1682.116,
    "Actual Total Time": 1682.118,
    "Actual Rows": 1,
    "Actual Loops": 1,
    "Shared Hit Blocks": 13392145,
    "Shared Read Blocks": 0,
    "Shared Dirtied Blocks": 0,
    "Shared Written Blocks": 0,
    "Local Hit Blocks": 0,
    "Local Read Blocks": 0,
    "Local Dirtied Blocks": 0,
    "Local Written Blocks": 0,
    "Temp Read Blocks": 0,
    "Temp Written Blocks": 0,
    "Plans": [
      {
        "Node Type": "Nested Loop",
        "Parent Relationship": "Outer",
        "Parallel Aware": false,
```

```
"Async Capable": false,
"Join Type": "Inner",
"Startup Cost": 8.08,
"Total Cost": 3279.51,
"Plan Rows": 290,
"Plan Width": 0,
"Actual Startup Time": 0.250,
"Actual Total Time": 1650.345,
"Actual Rows": 1067496,
"Actual Loops": 1,
"Inner Unique": false,
"Join Filter": "(t.id = ci.movie_id)",
"Rows Removed by Join Filter": 0,
"Shared Hit Blocks": 13392145,
"Shared Read Blocks": 0,
"Shared Dirtied Blocks": 0,
"Shared Written Blocks": 0,
"Local Hit Blocks": 0,
"Local Read Blocks": 0,
"Local Dirtied Blocks": 0,
"Local Written Blocks": 0,
"Temp Read Blocks": 0,
"Temp Written Blocks": 0,
"Plans": [
  {
    "Node Type": "Nested Loop",
    "Parent Relationship": "Outer",
    "Parallel Aware": false,
    "Async Capable": false,
    "Join Type": "Inner",
    "Startup Cost": 7.64,
```


"Total Cost": 2928.94,
"Plan Rows": 101,
"Plan Width": 16,
"Actual Startup Time": 0.108,
"Actual Total Time": 43.653,
"Actual Rows": 83888,
"Actual Loops": 1,
"Inner Unique": false,
"Join Filter": "(t.id = mi.movie_id)",
"Rows Removed by Join Filter": 0,
"Shared Hit Blocks": 66927,
"Shared Read Blocks": 0,
"Shared Dirtied Blocks": 0,
"Shared Written Blocks": 0,
"Local Hit Blocks": 0,
"Local Read Blocks": 0,
"Local Dirtied Blocks": 0,
"Local Written Blocks": 0,
"Temp Read Blocks": 0,
"Temp Written Blocks": 0,
"Plans": [
 {
 "Node Type": "Nested Loop",
 "Parent Relationship": "Outer",
 "Parallel Aware": false,
 "Async Capable": false,
 "Join Type": "Inner",
 "Startup Cost": 7.21,
 "Total Cost": 2790.22,
 "Plan Rows": 85,
 "Plan Width": 12,

"Actual Startup Time": 0.100,
"Actual Total Time": 5.071,
"Actual Rows": 2178,
"Actual Loops": 1,
"Inner Unique": true,
"Shared Hit Blocks": 14476,
"Shared Read Blocks": 0,
"Shared Dirtied Blocks": 0,
"Shared Written Blocks": 0,
"Local Hit Blocks": 0,
"Local Read Blocks": 0,
"Local Dirtied Blocks": 0,
"Local Written Blocks": 0,
"Temp Read Blocks": 0,
"Temp Written Blocks": 0,
"Plans": [
 {
 "Node Type": "Nested Loop",
 "Parent Relationship": "Outer",
 "Parallel Aware": false,
 "Async Capable": false,
 "Join Type": "Inner",
 "Startup Cost": 6.78,
 "Total Cost": 2044.75,
 "Plan Rows": 1438,
 "Plan Width": 8,
 "Actual Startup Time": 0.049,
 "Actual Total Time": 1.865,
 "Actual Rows": 3218,
 "Actual Loops": 1,
 "Inner Unique": false,

"Shared Hit Blocks": 1604,
"Shared Read Blocks": 0,
"Shared Dirtied Blocks": 0,
"Shared Written Blocks": 0,
"Local Hit Blocks": 0,
"Local Read Blocks": 0,
"Local Dirtied Blocks": 0,
"Local Written Blocks": 0,
"Temp Read Blocks": 0,
"Temp Written Blocks": 0,
"Plans": [
 {
 "Node Type": "Bitmap Heap Scan",
 "Parent Relationship": "Outer",
 "Parallel Aware": false,
 "Async Capable": false,
 "Relation Name": "movie_keyword",
 "Alias": "mk",
 "Startup Cost": 6.35,
 "Total Cost": 919.45,
 "Plan Rows": 247,
 "Plan Width": 4,
 "Actual Startup Time": 0.042,
 "Actual Total Time": 0.499,
 "Actual Rows": 414,
 "Actual Loops": 1,
 "Recheck Cond": "(keyword_id = 7084)",
 "Rows Removed by Index Recheck": 0,
 "Exact Heap Blocks": 358,
 "Lossy Heap Blocks": 0,
 "Shared Hit Blocks": 361,

```
"Shared Read Blocks": 0,
"Shared Dirtied Blocks": 0,
"Shared Written Blocks": 0,
"Local Hit Blocks": 0,
"Local Read Blocks": 0,
"Local Dirtied Blocks": 0,
"Local Written Blocks": 0,
"Temp Read Blocks": 0,
"Temp Written Blocks": 0,
"Plans": [
  {
    "Node Type": "Bitmap Index Scan",
    "Parent Relationship": "Outer",
    "Parallel Aware": false,
    "Async Capable": false,
    "Index Name": "keyword_id_movie_keyword",
    "Startup Cost": 0.00,
    "Total Cost": 6.29,
    "Plan Rows": 247,
    "Plan Width": 0,
    "Actual Startup Time": 0.020,
    "Actual Total Time": 0.021,
    "Actual Rows": 414,
    "Actual Loops": 1,
    "Index Cond": "(keyword_id = 7084)",
    "Shared Hit Blocks": 3,
    "Shared Read Blocks": 0,
    "Shared Dirtied Blocks": 0,
    "Shared Written Blocks": 0,
    "Local Hit Blocks": 0,
    "Local Read Blocks": 0,
```

```
"Local Dirtied Blocks": 0,  
"Local Written Blocks": 0,  
"Temp Read Blocks": 0,  
"Temp Written Blocks": 0  
}  
]  
,  
{  
  "Node Type": "Index Only Scan",  
  "Parent Relationship": "Inner",  
  "Parallel Aware": false,  
  "Async Capable": false,  
  "Scan Direction": "Forward",  
  "Index Name": "movie_id_movie_companies",  
  "Relation Name": "movie_companies",  
  "Alias": "mc",  
  "Startup Cost": 0.43,  
  "Total Cost": 4.50,  
  "Plan Rows": 6,  
  "Plan Width": 4,  
  "Actual Startup Time": 0.002,  
  "Actual Total Time": 0.003,  
  "Actual Rows": 8,  
  "Actual Loops": 414,  
  "Index Cond": "(movie_id = mk.movie_id)",  
  "Rows Removed by Index Recheck": 0,  
  "Heap Fetches": 0,  
  "Shared Hit Blocks": 1243,  
  "Shared Read Blocks": 0,  
  "Shared Dirtied Blocks": 0,  
  "Shared Written Blocks": 0,
```

```
"Local Hit Blocks": 0,  
"Local Read Blocks": 0,  
"Local Dirtied Blocks": 0,  
"Local Written Blocks": 0,  
"Temp Read Blocks": 0,  
"Temp Written Blocks": 0  
}  
]  
,  
{  
  "Node Type": "Index Scan",  
  "Parent Relationship": "Inner",  
  "Parallel Aware": false,  
  "Async Capable": false,  
  "Scan Direction": "Forward",  
  "Index Name": "title_pkey",  
  "Relation Name": "title",  
  "Alias": "t",  
  "Startup Cost": 0.43,  
  "Total Cost": 0.52,  
  "Plan Rows": 1,  
  "Plan Width": 4,  
  "Actual Startup Time": 0.001,  
  "Actual Total Time": 0.001,  
  "Actual Rows": 1,  
  "Actual Loops": 3218,  
  "Index Cond": "(id = mc.movie_id)",  
  "Rows Removed by Index Recheck": 0,  
  "Filter": "((production_year > 2000) AND (production_year < 2010))",  
  "Rows Removed by Filter": 0,  
  "Shared Hit Blocks": 12872,
```

```
"Shared Read Blocks": 0,
"Shared Dirtied Blocks": 0,
"Shared Written Blocks": 0,
"Local Hit Blocks": 0,
"Local Read Blocks": 0,
"Local Dirtied Blocks": 0,
"Local Written Blocks": 0,
"Temp Read Blocks": 0,
"Temp Written Blocks": 0
}
]
},
{
  "Node Type": "Index Scan",
  "Parent Relationship": "Inner",
  "Parallel Aware": false,
  "Async Capable": false,
  "Scan Direction": "Forward",
  "Index Name": "movie_id_movie_info",
  "Relation Name": "movie_info",
  "Alias": "mi",
  "Startup Cost": 0.43,
  "Total Cost": 1.53,
  "Plan Rows": 8,
  "Plan Width": 4,
  "Actual Startup Time": 0.002,
  "Actual Total Time": 0.015,
  "Actual Rows": 39,
  "Actual Loops": 2178,
  "Index Cond": "(movie_id = mc.movie_id)",
  "Rows Removed by Index Recheck": 0,
```

```
"Filter": "(info_type_id = 16)",
"Rows Removed by Filter": 175,
"Shared Hit Blocks": 52451,
"Shared Read Blocks": 0,
"Shared Dirtied Blocks": 0,
"Shared Written Blocks": 0,
"Local Hit Blocks": 0,
"Local Read Blocks": 0,
"Local Dirtied Blocks": 0,
"Local Written Blocks": 0,
"Temp Read Blocks": 0,
"Temp Written Blocks": 0
}
],
{
  "Node Type": "Index Scan",
  "Parent Relationship": "Inner",
  "Parallel Aware": false,
  "Async Capable": false,
  "Scan Direction": "Forward",
  "Index Name": "movie_id_cast_info",
  "Relation Name": "cast_info",
  "Alias": "ci",
  "Startup Cost": 0.44,
  "Total Cost": 3.38,
  "Plan Rows": 7,
  "Plan Width": 4,
  "Actual Startup Time": 0.004,
  "Actual Total Time": 0.018,
  "Actual Rows": 13,
```



```
"Actual Loops": 83888,
"Index Cond": "(movie_id = mi.movie_id)",
"Rows Removed by Index Recheck": 0,
"Filter": "(role_id = 2)",
"Rows Removed by Filter": 152,
"Shared Hit Blocks": 13325218,
"Shared Read Blocks": 0,
"Shared Dirtied Blocks": 0,
"Shared Written Blocks": 0,
"Local Hit Blocks": 0,
"Local Read Blocks": 0,
"Local Dirtied Blocks": 0,
"Local Written Blocks": 0,
"Temp Read Blocks": 0,
"Temp Written Blocks": 0
}
]
}
],
"Planning": {
  "Shared Hit Blocks": 160,
  "Shared Read Blocks": 0,
  "Shared Dirtied Blocks": 0,
  "Shared Written Blocks": 0,
  "Local Hit Blocks": 0,
  "Local Read Blocks": 0,
  "Local Dirtied Blocks": 0,
  "Local Written Blocks": 0,
  "Temp Read Blocks": 0,
  "Temp Written Blocks": 0
}
```

```
},  
  "Planning Time": 0.902,  
  "Triggers": [  
  ],  
  "Execution Time": 1682.153  
}  
]
```

Execution plan after optimization in JSON format:

```
[  
  {  
    "Plan": {  
      "Node Type": "Aggregate",  
      "Strategy": "Plain",  
      "Partial Mode": "Simple",  
      "Parallel Aware": false,  
      "Async Capable": false,  
      "Startup Cost": 3876.25,  
      "Total Cost": 3876.26,  
      "Plan Rows": 1,  
      "Plan Width": 8,  
      "Actual Startup Time": 232.139,  
      "Actual Total Time": 232.141,  
      "Actual Rows": 1,  
      "Actual Loops": 1,  
      "Shared Hit Blocks": 166857,  
      "Shared Read Blocks": 339,  
      "Shared Dirtied Blocks": 0,  
      "Shared Written Blocks": 0,  
      "Local Hit Blocks": 0,
```

```
"Local Read Blocks": 0,
"Local Dirtied Blocks": 0,
"Local Written Blocks": 0,
"Temp Read Blocks": 0,
"Temp Written Blocks": 0,
"Plans": [
  {
    "Node Type": "Nested Loop",
    "Parent Relationship": "Outer",
    "Parallel Aware": false,
    "Async Capable": false,
    "Join Type": "Inner",
    "Startup Cost": 8.08,
    "Total Cost": 3875.52,
    "Plan Rows": 291,
    "Plan Width": 0,
    "Actual Startup Time": 6.768,
    "Actual Total Time": 204.579,
    "Actual Rows": 1067496,
    "Actual Loops": 1,
    "Inner Unique": false,
    "Join Filter": "(t.id = mc.movie_id)",
    "Rows Removed by Join Filter": 0,
    "Shared Hit Blocks": 166857,
    "Shared Read Blocks": 339,
    "Shared Dirtied Blocks": 0,
    "Shared Written Blocks": 0,
    "Local Hit Blocks": 0,
    "Local Read Blocks": 0,
    "Local Dirtied Blocks": 0,
    "Local Written Blocks": 0,
```

```
"Temp Read Blocks": 0,
"Temp Written Blocks": 0,
"Plans": [
{
  "Node Type": "Nested Loop",
  "Parent Relationship": "Outer",
  "Parallel Aware": false,
  "Async Capable": false,
  "Join Type": "Inner",
  "Startup Cost": 7.65,
  "Total Cost": 3702.70,
  "Plan Rows": 282,
  "Plan Width": 16,
  "Actual Startup Time": 4.391,
  "Actual Total Time": 91.866,
  "Actual Rows": 41636,
  "Actual Loops": 1,
  "Inner Unique": false,
  "Join Filter": "(t.id = mi.movie_id)",
  "Rows Removed by Join Filter": 0,
  "Shared Hit Blocks": 41948,
  "Shared Read Blocks": 339,
  "Shared Dirtied Blocks": 0,
  "Shared Written Blocks": 0,
  "Local Hit Blocks": 0,
  "Local Read Blocks": 0,
  "Local Dirtied Blocks": 0,
  "Local Written Blocks": 0,
  "Temp Read Blocks": 0,
  "Temp Written Blocks": 0,
  "Plans": [
```

```
{
  "Node Type": "Nested Loop",
  "Parent Relationship": "Outer",
  "Parallel Aware": false,
  "Async Capable": false,
  "Join Type": "Inner",
  "Startup Cost": 7.22,
  "Total Cost": 3356.52,
  "Plan Rows": 238,
  "Plan Width": 12,
  "Actual Startup Time": 4.004,
  "Actual Total Time": 67.304,
  "Actual Rows": 1254,
  "Actual Loops": 1,
  "Inner Unique": false,
  "Shared Hit Blocks": 13672,
  "Shared Read Blocks": 311,
  "Shared Dirtied Blocks": 0,
  "Shared Written Blocks": 0,
  "Local Hit Blocks": 0,
  "Local Read Blocks": 0,
  "Local Dirtied Blocks": 0,
  "Local Written Blocks": 0,
  "Temp Read Blocks": 0,
  "Temp Written Blocks": 0,
  "Plans": [
    {
      "Node Type": "Nested Loop",
      "Parent Relationship": "Outer",
      "Parallel Aware": false,
      "Async Capable": false,
```

"Join Type": "Inner",
"Startup Cost": 6.78,
"Total Cost": 2991.22,
"Plan Rows": 83,
"Plan Width": 8,
"Actual Startup Time": 0.067,
"Actual Total Time": 20.450,
"Actual Rows": 199,
"Actual Loops": 1,
"Inner Unique": true,
"Shared Hit Blocks": 1922,
"Shared Read Blocks": 95,
"Shared Dirtied Blocks": 0,
"Shared Written Blocks": 0,
"Local Hit Blocks": 0,
"Local Read Blocks": 0,
"Local Dirtied Blocks": 0,
"Local Written Blocks": 0,
"Temp Read Blocks": 0,
"Temp Written Blocks": 0,
"Plans": [
 {
 "Node Type": "Bitmap Heap Scan",
 "Parent Relationship": "Outer",
 "Parallel Aware": false,
 "Async Capable": false,
 "Relation Name": "movie_keyword",
 "Alias": "mk",
 "Startup Cost": 6.35,
 "Total Cost": 919.45,
 "Plan Rows": 247,

"Plan Width": 4,
"Actual Startup Time": 0.042,
"Actual Total Time": 0.469,
"Actual Rows": 414,
"Actual Loops": 1,
"Recheck Cond": "(keyword_id = 7084)",
"Rows Removed by Index Recheck": 0,
"Exact Heap Blocks": 358,
"Lossy Heap Blocks": 0,
"Shared Hit Blocks": 361,
"Shared Read Blocks": 0,
"Shared Dirtied Blocks": 0,
"Shared Written Blocks": 0,
"Local Hit Blocks": 0,
"Local Read Blocks": 0,
"Local Dirtied Blocks": 0,
"Local Written Blocks": 0,
"Temp Read Blocks": 0,
"Temp Written Blocks": 0,
"Plans": [
 {
 "Node Type": "Bitmap Index Scan",
 "Parent Relationship": "Outer",
 "Parallel Aware": false,
 "Async Capable": false,
 "Index Name": "keyword_id_movie_keyword",
 "Startup Cost": 0.00,
 "Total Cost": 6.29,
 "Plan Rows": 247,
 "Plan Width": 0,
 "Actual Startup Time": 0.021,

```
"Actual Total Time": 0.021,  
"Actual Rows": 414,  
"Actual Loops": 1,  
"Index Cond": "(keyword_id = 7084)",  
"Shared Hit Blocks": 3,  
"Shared Read Blocks": 0,  
"Shared Dirtied Blocks": 0,  
"Shared Written Blocks": 0,  
"Local Hit Blocks": 0,  
"Local Read Blocks": 0,  
"Local Dirtied Blocks": 0,  
"Local Written Blocks": 0,  
"Temp Read Blocks": 0,  
"Temp Written Blocks": 0  
}  
],  
{  
  "Node Type": "Index Scan",  
  "Parent Relationship": "Inner",  
  "Parallel Aware": false,  
  "Async Capable": false,  
  "Scan Direction": "Forward",  
  "Index Name": "title_pkey",  
  "Relation Name": "title",  
  "Alias": "t",  
  "Startup Cost": 0.43,  
  "Total Cost": 8.39,  
  "Plan Rows": 1,  
  "Plan Width": 4,  
  "Actual Startup Time": 0.048,
```



```
"Actual Total Time": 0.048,  
"Actual Rows": 0,  
"Actual Loops": 414,  
"Index Cond": "(id = mk.movie_id)",  
"Rows Removed by Index Recheck": 0,  
"Filter": "((production_year > 2000) AND (production_year < 2010))",  
"Rows Removed by Filter": 1,  
"Shared Hit Blocks": 1561,  
"Shared Read Blocks": 95,  
"Shared Dirtied Blocks": 0,  
"Shared Written Blocks": 0,  
"Local Hit Blocks": 0,  
"Local Read Blocks": 0,  
"Local Dirtied Blocks": 0,  
"Local Written Blocks": 0,  
"Temp Read Blocks": 0,  
"Temp Written Blocks": 0  
}  
]  
,  
{  
  "Node Type": "Index Scan",  
  "Parent Relationship": "Inner",  
  "Parallel Aware": false,  
  "Async Capable": false,  
  "Scan Direction": "Forward",  
  "Index Name": "movie_id_cast_info",  
  "Relation Name": "cast_info",  
  "Alias": "ci",  
  "Startup Cost": 0.44,  
  "Total Cost": 4.33,
```

```
"Plan Rows": 7,  
"Plan Width": 4,  
"Actual Startup Time": 0.157,  
"Actual Total Time": 0.235,  
"Actual Rows": 6,  
"Actual Loops": 199,  
"Index Cond": "(movie_id = t.id)",  
"Rows Removed by Index Recheck": 0,  
"Filter": "(role_id = 2)",  
"Rows Removed by Filter": 55,  
"Shared Hit Blocks": 11750,  
"Shared Read Blocks": 216,  
"Shared Dirtied Blocks": 0,  
"Shared Written Blocks": 0,  
"Local Hit Blocks": 0,  
"Local Read Blocks": 0,  
"Local Dirtied Blocks": 0,  
"Local Written Blocks": 0,  
"Temp Read Blocks": 0,  
"Temp Written Blocks": 0  
}  
}  
},  
{  
  "Node Type": "Index Scan",  
  "Parent Relationship": "Inner",  
  "Parallel Aware": false,  
  "Async Capable": false,  
  "Scan Direction": "Forward",  
  "Index Name": "movie_id_movie_info",  
  "Relation Name": "movie_info",
```

```
"Alias": "mi",
"Startup Cost": 0.43,
"Total Cost": 1.35,
"Plan Rows": 8,
"Plan Width": 4,
"Actual Startup Time": 0.005,
"Actual Total Time": 0.017,
"Actual Rows": 33,
"Actual Loops": 1254,
"Index Cond": "(movie_id = ci.movie_id)",
"Rows Removed by Index Recheck": 0,
"Filter": "(info_type_id = 16)",
"Rows Removed by Filter": 150,
"Shared Hit Blocks": 28276,
"Shared Read Blocks": 28,
"Shared Dirtied Blocks": 0,
"Shared Written Blocks": 0,
"Local Hit Blocks": 0,
"Local Read Blocks": 0,
"Local Dirtied Blocks": 0,
"Local Written Blocks": 0,
"Temp Read Blocks": 0,
"Temp Written Blocks": 0
}
]
},
{
  "Node Type": "Index Only Scan",
  "Parent Relationship": "Inner",
  "Parallel Aware": false,
  "Async Capable": false,
```

```
"Scan Direction": "Forward",
"Index Name": "movie_id_movie_companies",
"Relation Name": "movie_companies",
"Alias": "mc",
"Startup Cost": 0.43,
"Total Cost": 0.54,
"Plan Rows": 6,
"Plan Width": 4,
"Actual Startup Time": 0.000,
"Actual Total Time": 0.001,
"Actual Rows": 26,
"Actual Loops": 41636,
"Index Cond": "(movie_id = ci.movie_id)",
"Rows Removed by Index Recheck": 0,
"Heap Fetches": 0,
"Shared Hit Blocks": 124909,
"Shared Read Blocks": 0,
"Shared Dirtied Blocks": 0,
"Shared Written Blocks": 0,
"Local Hit Blocks": 0,
"Local Read Blocks": 0,
"Local Dirtied Blocks": 0,
"Local Written Blocks": 0,
"Temp Read Blocks": 0,
"Temp Written Blocks": 0
}
]
}
]
},
"Planning": {
```

```
"Shared Hit Blocks": 64,  
"Shared Read Blocks": 0,  
"Shared Dirtied Blocks": 0,  
"Shared Written Blocks": 0,  
"Local Hit Blocks": 0,  
"Local Read Blocks": 0,  
"Local Dirtied Blocks": 0,  
"Local Written Blocks": 0,  
"Temp Read Blocks": 0,  
"Temp Written Blocks": 0  
},  
"Planning Time": 0.344,  
"Triggers": [  
],  
"Execution Time": 232.171  
}  
]
```